

LAB 1.2: Construction and Testing of an RJ-45 Ethernet Cable

Objective:

To construct an RJ-45 Ethernet cable using UTP cable and RJ-45 connectors.

Theory

An RJ-45 cable is a widely used networking cable employed in Ethernet-based Local Area Networks (LANs). The term RJ-45 (Registered Jack-45) refers to the standardized 8-pin modular connector used to terminate twisted-pair cables. RJ-45 cables provide a reliable medium for data transmission between networking devices such as computers, switches, routers, hubs, and access points.

The construction of an RJ-45 cable involves arranging the conductors of a twisted-pair cable according to a specified wiring standard and terminating them using RJ-45 connectors. Proper construction and testing ensure reliable communication, reduced signal interference, and compliance with Ethernet networking standards.

Common Network Cable Types

Network communication utilizes different types of transmission media depending on bandwidth requirements, distance, and environmental conditions.

1. Twisted Pair Cable

Twisted pair cable consists of insulated copper wires twisted together to minimize electromagnetic interference (EMI) and crosstalk. It is the most commonly used cable in Ethernet networks.

Types of twisted pair cables include:

- **UTP (Unshielded Twisted Pair):** Does not contain additional shielding. It is inexpensive, flexible, and widely used in LANs.
- **STP (Shielded Twisted Pair):** Contains metallic shielding to provide greater protection against electromagnetic interference.

2. Coaxial Cable

Coaxial cable consists of a central conductor surrounded by insulation, shielding, and an outer protective jacket. It was widely used in earlier Ethernet implementations and is still used in cable television and broadband networks.

3. Fiber Optic Cable

Fiber optic cable transmits data using light signals through glass or plastic fibers. It offers high bandwidth, low attenuation, and immunity to electromagnetic interference, making it suitable for long-distance and high-speed communication.

UTP Cable Categories

UTP cables are classified into categories based on their transmission capabilities and performance.

Category	Maximum Speed	Bandwidth
Cat 3	10 Mbps	16 MHz
Cat 5	100 Mbps	100 MHz
Cat 5e	1 Gbps	100 MHz
Cat 6	1–10 Gbps	250 MHz
Cat 6a	10 Gbps	500 MHz
Cat 7	10 Gbps	600 MHz
Cat 8	25–40 Gbps	2000 MHz

Cat 5e and Cat 6 cables are the most commonly used categories in modern Ethernet networks due to their balance of cost and performance.

RJ-45 Connector and Pin Configuration

The RJ-45 connector contains eight pins corresponding to the eight conductors of a twisted-pair cable. Ethernet cabling follows standardized wiring schemes defined by the TIA/EIA standards:

T568B Standard

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Straight-Through Cable

A straight-through cable is created by terminating both ends of the cable using the same wiring standard, either T568A-T568A or T568B-T568B.

Applications

- Computer to Switch
- Computer to Hub
- Router to Switch
- Access Point to Switch

Straight-through cables are the most commonly used Ethernet cables in modern LAN installations.

Cross-Over Cable

A cross-over cable is constructed by terminating one end using the T568A standard and the other end using the T568B standard.

In this cable, the transmit and receive pairs are crossed so that similar devices can communicate directly.

Applications

- Computer to Computer
- Switch to Switch
- Hub to Hub
- Router to Router

Components Required

1. Cat5e UTP Cable
2. RJ-45 Connectors
3. Cable Stripper
4. Scissors
5. Crimping Tool
6. Cable Tester

Procedure:

1.2.1) Making the Connection (Construction of RJ-45 Cable)

Step 1: Strip the Cable End

1. Strip approximately 1 to 1.5 inches of the outer insulating sheath from the end of the UTP cable using a cable stripper.

Step 2: Untwist the Wire Ends

1. Separate and untwist the four twisted pairs of wires.
2. Arrange the conductors according to their insulation colors.

Step 3: Arrange the Wires

Arrange the wires according to the required wiring standard.

TIA/EIA 568B Standard

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Step 4: Trim the Wires to Size

1. Straighten the conductors and trim them evenly.
2. Leave approximately 0.5 inch of exposed conductors beyond the cable sheath.

Step 5: Attach the RJ-45 Connector

1. Hold the RJ-45 connector with the locking tab facing downward.
2. Insert the wires into the connector while maintaining the correct wire sequence from left to right.
3. Ensure that each conductor enters its respective channel inside the connector.

Step 6: Verify the Arrangement

1. Confirm that all wires extend fully to the end of the connector.
2. Verify that the wire sequence is correct.
3. Ensure that the outer cable sheath is inserted sufficiently inside the connector for proper strain relief.

Step 7: Crimp the Connector

1. Insert the connector into the crimping tool.
2. Squeeze the handles firmly to crimp the connector onto the cable.
3. Repeat the same process for the opposite end of the cable according to the required cable type (straight-through or cross-over).

1.2.2) Testing the Cable**Step 8: Test the RJ-45 Cable**

1. Connect both ends of the cable to a cable tester.
2. Power on the tester and observe the pin-to-pin connectivity results.
3. Verify that all eight conductors are properly connected.
4. Record the test results and confirm whether the cable is functioning correctly.

Output

A functional Ethernet patch cable was successfully fabricated and tested. The cable tester verified proper pin-to-pin continuity and confirmed that the cable was correctly terminated according to the selected Ethernet wiring standard.



Fig i) RJ-45 Cable

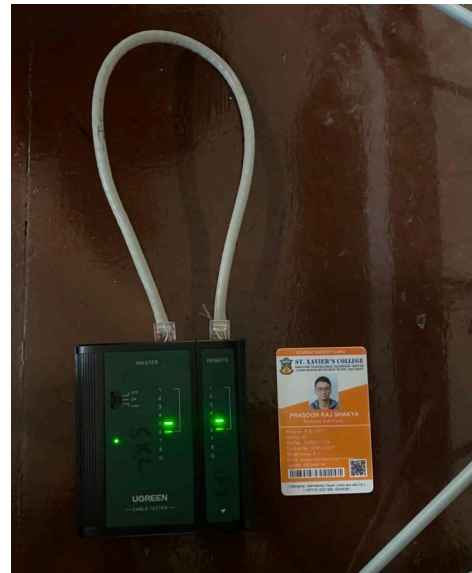


Fig ii) Testing of RJ-45 Cable

Conclusion

The RJ-45 Ethernet cable was successfully constructed by terminating a UTP cable according to the specified wiring standard. The cable was tested using a cable tester, and proper pin-to-pin connectivity was verified. This experiment provided practical knowledge of RJ-45 cable construction, T568A/T568B wiring standards, and the testing procedures used to ensure reliable Ethernet network communication.